

Radiation, Magnitudes & Spectra

when a cosmic ray splits you in half but you're too lazy to die

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Prerequisite Knowledge

- The ideal gas law, $PV = nRT$, and its two special cases: Boyle's Law and Charles's Law
- Calorimetry: $Q = mc\Delta T$
- Basic algebra, logarithms, and single-variable calculus
- Braincells

1 Blackbody Radiation, From First Principles

1.1 The Idealized Blackbody

Real objects are messy: shiny metal reflects most light instead of absorbing it, and a white wall reflects visible light much more than a black one. To strip away all that material-specific complexity, physicists imagine an idealized **blackbody**: an object that absorbs *every* wavelength of light that falls on it, reflecting nothing (hence “black”).

Here’s the useful trick: a perfect absorber must also be a perfect *emitter*. If it weren’t — if it absorbed everything but emitted less efficiently than it absorbed — it would keep heating up forever with no way to shed that energy, which can’t happen for an object sitting in equilibrium with its surroundings. So a blackbody’s own glow, at a given temperature, is the maximum possible emission any object can produce at that temperature.

Stars are, to amazing approximation, blackbodies. Even though a star is made of gas, not a solid black surface, the gas is so dense and opaque that light bounces around and is absorbed and re-emitted an enormous number of times before finally escaping from the surface. By the time it escapes, the radiation has completely “forgotten” any details of how it started and settled into the universal blackbody shape, fixed entirely by temperature.

1.2 The Blackbody Spectrum: One Curve, One Number

A blackbody doesn’t radiate all wavelengths equally. If you measure how much power comes out at each wavelength, you get a smooth, single-humped curve whose exact shape depends on nothing except temperature.

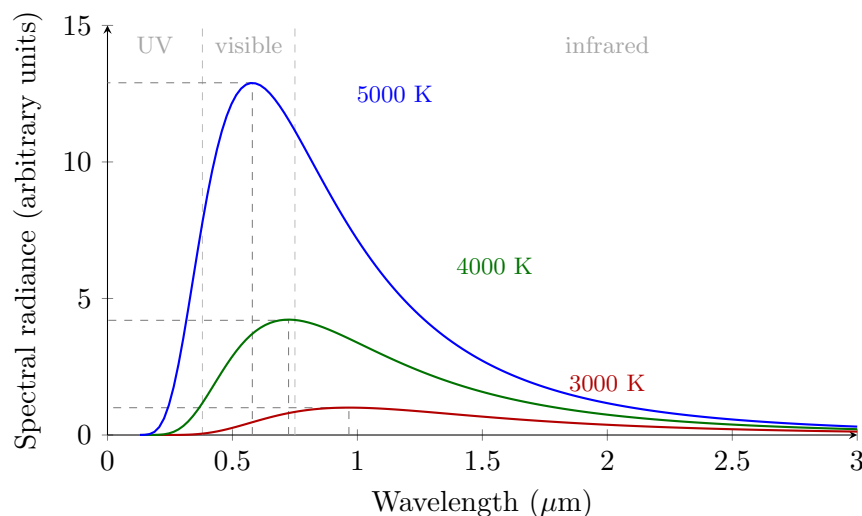


Figure 1: Blackbody curves at three temperatures. Two things to notice, and they are the entire content of Sections 1.3 and 1.4: the peak visibly slides to shorter wavelength as temperature rises (dashed guides) — this is **Wien’s Law** — and the curves don’t just shift, they grow dramatically taller and wider too, since the total area under a curve (the total power radiated) grows enormously with temperature — this is **Stefan’s Law**. A single number, T , fixes the entire curve.

This is why a heated piece of metal glows red, then orange, then white, then bluish-white as it gets hotter: it isn’t switching colors arbitrarily, it’s sliding along these exact curves.

1.3 Wien's Displacement Law

The wavelength at which the curve peaks, λ_{max} , is inversely proportional to temperature:

$$\lambda_{max}T = b, \quad b \approx 2.898 \times 10^{-3} \text{ m K.}$$

Hotter objects peak at shorter wavelengths — bluer light is more energetic, and a hotter object has more energy to hand out per photon. This single formula is why color is a thermometer: measure where a star's spectrum peaks, and you immediately know its surface temperature, literally no other information required.

Worked Example

The Sun's spectrum peaks at about $\lambda_{max} = 502 \text{ nm}$ (blue-green). Estimate its surface temperature.

$$T = \frac{b}{\lambda_{max}} = \frac{2.898 \times 10^{-3}}{502 \times 10^{-9}} \approx 5775 \text{ K,}$$

matching the Sun's known effective temperature almost exactly.

1.4 Stefan's Law

The *height and width* of the curve — not just where it peaks — also depend on temperature, and much more dramatically. The total power radiated per unit surface area, E , is the total area under the curve (summing up the contribution from every wavelength), and this area turns out to scale as the *fourth power* of temperature:

$$E = \sigma T^4, \quad \sigma \approx 5.67 \times 10^{-8} \text{ W m}^{-2}\text{K}^{-4}.$$

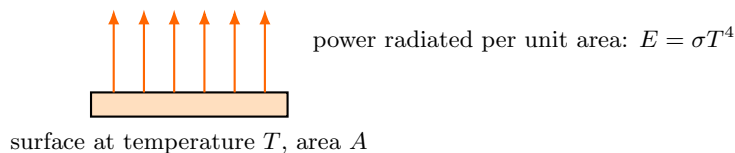


Figure 2: Every patch of a blackbody's surface radiates power outward according to Stefan's Law, regardless of the direction it happens to face.

The fourth-power dependence is extreme: double an object's temperature, and it radiates *sixteen* times more power per unit area. This single fact is why small changes in temperature (a star contracting slightly, a planet's greenhouse gases trapping a bit more heat) can have outsized effects on total energy output.

1.5 From a Point on a Star's Surface to Its Total Luminosity

Stefan's Law gives the power radiated by *one square meter* of surface. To get a star's **total luminosity** L — the total power radiated in every direction, from its entire surface — multiply by the star's whole surface area, $4\pi R^2$:

$$L = 4\pi R^2 \sigma T^4.$$

Nothing new is happening physically; this is just Stefan's Law scaled up from “per unit area” to “the whole sphere.” But it's a very very very very useful equation, because L , R , and T are three

of the most fundamental numbers describing any star, and this formula ties them together: know any two, and the third pops out immediately.

1.6 Flux, Distance, and the Inverse Square Law

Luminosity is what a star *emits*. What we actually *measure* on Earth is much smaller, because that light has spread out over an enormous sphere by the time it reaches us.

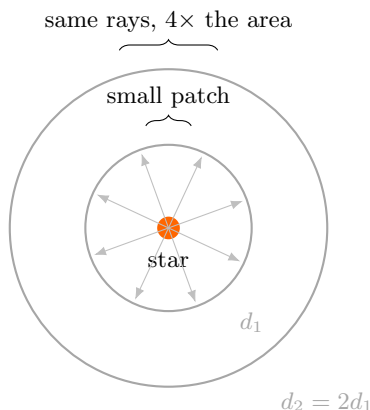


Figure 3: The same bundle of light rays passes through a patch of area A on the inner sphere (radius d_1) and a patch of area $4A$ on the outer sphere (radius $d_2 = 2d_1$) — same total energy, four times the area, so a quarter of the flux. This is the inverse square law.

The total power L is spread evenly over a sphere of radius d (the distance to the star), whose surface area is $4\pi d^2$. The power received per unit area — the **flux** f — is therefore

$$f = \frac{L}{4\pi d^2}.$$

Double the distance, and the same light is smeared over four times the area, so the flux drops by a factor of four — an **inverse square law**, identical in spirit to gravity or Coulomb’s law diluting over a sphere. Combining this with Section 1.5’s luminosity formula lets you connect what a telescope actually measures (f) to a star’s physical size and temperature (R , T), provided you know the distance d .

Worked Example

A star has surface temperature $T = 6000$ K and radius $R = 1.2R_\odot$, and sits at distance $d = 10$ pc. Find the flux received at Earth.

First, $L = 4\pi R^2 \sigma T^4$. Using $R_\odot \approx 6.96 \times 10^8$ m, so $R \approx 8.35 \times 10^8$ m:

$$L = 4\pi(8.35 \times 10^8)^2(5.67 \times 10^{-8})(6000)^4 \approx 1.62 \times 10^{26} \text{ W}.$$

Then with $d = 10$ pc $\approx 3.09 \times 10^{17}$ m:

$$f = \frac{L}{4\pi d^2} \approx \frac{1.62 \times 10^{26}}{4\pi(3.09 \times 10^{17})^2} \approx 1.35 \times 10^{-9} \text{ W/m}^2.$$

Notice how the boxed formulas of this section are the *entire* calculation — Stefan’s Law, scaled to a sphere, diluted by distance. No new physics anywhere in this chain. It’s that easy.

2 The Magnitude System

2.1 Apparent Magnitude and Pogson’s Equation

Astronomers don’t usually quote flux directly, for a historical reason: over two thousand years ago, Hipparchus ranked stars by eye on a scale of 1 (brightest) to 6 (faintest). This stuck around, but the human eye doesn’t perceive brightness linearly — it perceives it *logarithmically* (the same reason decibels and pH are logarithmic scales). In 1856, Norman Pogson formalized this into a precise rule: a difference of 5 magnitudes corresponds to *exactly* a factor of 100 in flux. This gives **Pogson’s equation**:

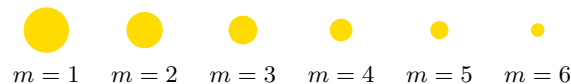
$$m_1 - m_2 = -2.5 \log_{10} \frac{f_1}{f_2}.$$


Figure 4: Each step of one magnitude corresponds to a fixed ratio of brightness ($100^{1/5} \approx 2.512\times$), not a fixed difference — which is exactly why the dots above shrink by the same factor each time, not the same amount. Smaller magnitude number means brighter; this is a famously backwards convention, but it’s the one everyone still uses.

The negative sign is what makes bigger flux correspond to a *smaller* magnitude number — counterintuitive at first, but it’s what preserves Hipparchus’s original “1 is brightest” convention.

2.2 Absolute Magnitude and the Distance Modulus

Apparent magnitude m mixes together two completely different things: how intrinsically luminous a star is, and how far away it happens to be. A dim, nearby star and a brilliant, distant one can have the exact same m . To separate these, astronomers define the **absolute magnitude** M : the apparent magnitude a star *would* have if it were moved to a standard distance of exactly 10 pc.

Since $f \propto L/d^2$ (established above), plugging two flux ratios — the real one at distance d , and the hypothetical one at 10 pc — into Pogson’s equation gives

$$m - M = -2.5 \log_{10} \frac{L/4\pi d^2}{L/4\pi(10\text{ pc})^2} = 5 \log_{10} \frac{d}{10\text{ pc}}.$$

$$m - M = 5 \log_{10} \left(\frac{d}{10\text{ pc}} \right) \equiv \text{DM},$$

where DM is called the **distance modulus**. Notice the luminosity L cancelled out entirely — the distance modulus depends *only* on distance, which is exactly why it’s useful: measure m with a telescope, know (or look up) M from a star’s type, and d falls right out. This single rearranged equation is the workhorse behind most distance measurements in astronomy that don’t rely on parallax.

Worked Example

A star has apparent magnitude $m = 8.2$ and absolute magnitude $M = 1.4$. How far away is

it?

$$m - M = 6.8 = 5 \log_{10} \left(\frac{d}{10 \text{ pc}} \right) \Rightarrow \frac{d}{10 \text{ pc}} = 10^{6.8/5} \approx 22.9 \Rightarrow d \approx 229 \text{ pc}.$$

3 Spectroscopy: Reading a Star's Fingerprint

3.1 Three Kinds of Spectra

Photometry (Sections 1–2) tells you *how much* light a star sends us, but not what wavelengths that light is made of. Spreading starlight out into its component colors — a **spectrum** — reveals far more: what elements the star (or the gas between us and it) is made of, and even how fast it's moving.

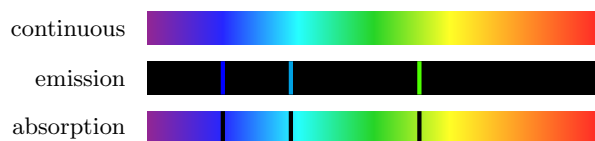


Figure 5: A dense hot object (like a star's interior) emits a continuous rainbow. A hot, thin gas glows only at specific wavelengths — an **emission** spectrum. The same gas, when cooler and lying in front of a continuous source, instead removes light at those same wavelengths — an **absorption** spectrum. Most stellar spectra are of this third kind: a continuous blackbody spectrum from the star's interior, with dark absorption lines carved out by the cooler gas in its outer atmosphere.

3.2 The Bohr Model of the Atom

Why does a gas absorb or emit only *specific* wavelengths, rather than a smooth spread? The answer is that an electron bound to an atom cannot orbit at just any energy — only certain discrete orbits, and therefore discrete energies, are allowed. This is the essential idea behind the (simplified) **Bohr model** of hydrogen: an electron circles the nucleus only in orbits where its angular momentum is an integer multiple of $\hbar = h/2\pi$, which forces its energy to take one of a discrete ladder of values, $E_n \propto -1/n^2$, labeled by a positive integer n called the **principal quantum number**.

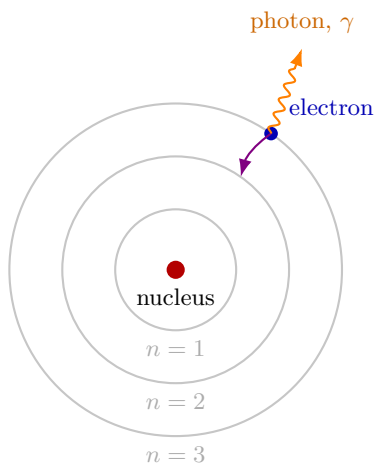


Figure 6: An electron dropping from a higher orbit ($n = 3$) to a lower one ($n = 2$) — violet arrow

— must shed the energy difference $\Delta E = E_3 - E_2$ somewhere, and it does so by emitting a single photon of exactly that energy (orange wavy arrow). The reverse process — an electron absorbing a photon of exactly the right energy to jump up an orbit — is what carves out absorption lines. Because only discrete ΔE 's are available, only discrete wavelengths ($E = hc/\lambda$) appear, which is the entire reason line spectra exist at all.

3.3 The Rydberg Formula and Spectral Series

Since $E_n \propto -1/n^2$, the energy released dropping from level n_2 to level n_1 (with $n_2 > n_1$) is $\Delta E \propto \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$. Writing this out in terms of wavelength via $\Delta E = hc/\lambda$ gives the **Rydberg formula**:

$$\frac{1}{\lambda} = R_{\infty} Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right),$$

where Z is the nuclear charge (1 for hydrogen) and $R_{\infty} \approx 1.097 \times 10^7 \text{ m}^{-1}$ is the **Rydberg constant**. Transitions landing on the same final level n_1 form a named **series**:

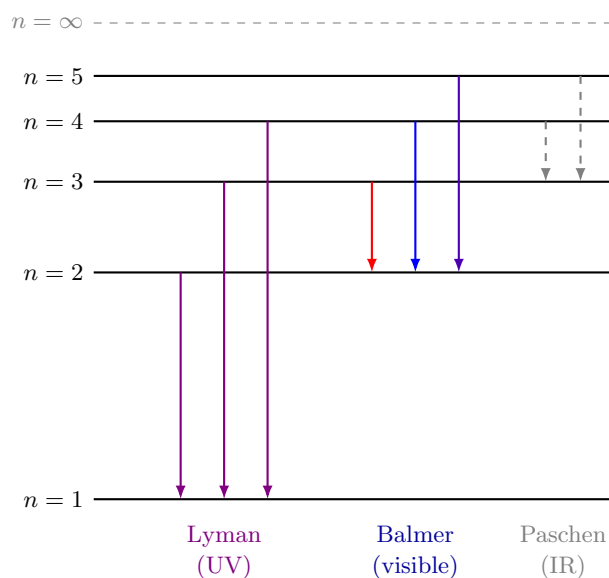


Figure 7: Energy levels bunch closer together as n increases (matching $E_n \propto -1/n^2$), which is why the transition wavelengths within a series get closer together too. The **Lyman series** (landing on $n=1$) is energetic enough to fall in the ultraviolet; the **Balmer series** (landing on $n=2$) is the one that happens to fall in visible light, which is why it's the series you actually see in a star's optical spectrum; the **Paschen series** (landing on $n=3$) falls in the infrared.

3.4 Spectral Classification and the Hertzsprung–Russell Diagram

A star's absorption-line pattern depends almost entirely on its surface temperature (hotter gas ionizes differently than cooler gas), which is why stars are sorted into temperature classes, traditionally remembered by the mnemonic “*Oh Be A Fine Girl/Guy, Kiss Me*”:

Class	Approx. Temperature (K)	Color
O	> 30,000	blue
B	10,000–30,000	blue-white
A	7,300–10,000	white
F	6,000–7,300	yellow-white
G	5,300–6,000	yellow (the Sun is G2)
K	3,900–5,300	orange
M	< 3,900	red

Plotting luminosity against temperature (temperature axis conventionally *decreasing* to the right) produces the single most important diagram in stellar astrophysics:

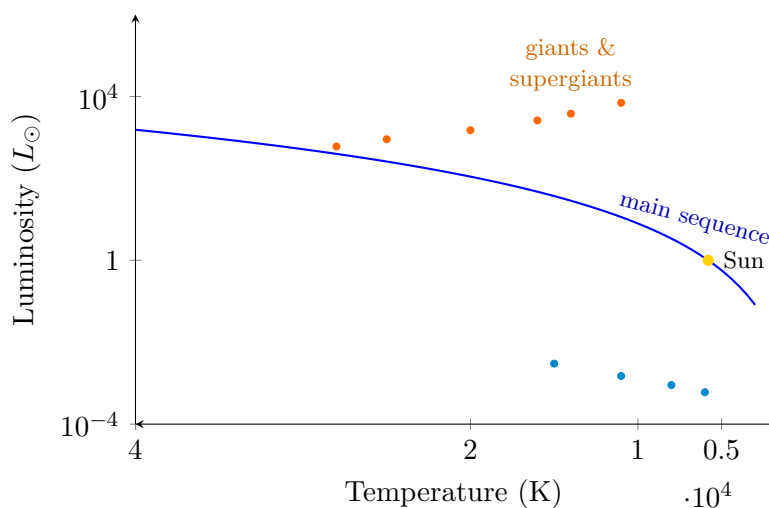


Figure 8: The Hertzsprung–Russell (HR) diagram. Most stars, including the Sun, sit along the diagonal **main sequence** — the band where stars spend the vast majority of their lives steadily fusing hydrogen. Stars that have swollen up late in life sit in the **giants/supergiants** region (cool but enormous, hence luminous); collapsed stellar remnants sit among the **white dwarfs** (hot but tiny, hence dim).

3.5 The Mass-Luminosity Relation

For main-sequence stars specifically, luminosity climbs steeply with mass:

$$\frac{L}{L_{\odot}} = \left(\frac{M}{M_{\odot}} \right)^{3.5}.$$

This is an empirical law (found by measurement, not derived from first principles here), but it has a real consequence that you want to remember: a more massive star burns through its fuel supply *much* faster than the extra fuel it has to burn, so massive stars live dramatically *shorter* lives than low-mass stars, despite starting with more fuel in the tank.

3.6 The Doppler Effect and Redshift

Everything above assumes a star sitting still relative to us. If it's moving along our line of sight, its spectral lines shift.

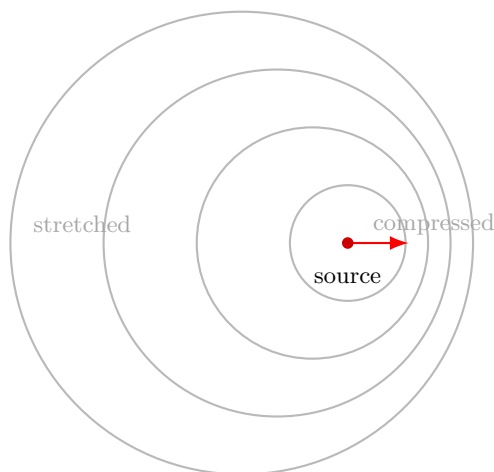


Figure 9: A moving source emits wavefronts (circles) from a slightly different position each time. Ahead of its motion (right), successive wavefronts pile up — shorter wavelength, a **blueshift**. Behind it (left), wavefronts spread apart — longer wavelength, a **redshift**.

We define the **redshift** z as the fractional change in wavelength, $z = (\lambda - \lambda_0)/\lambda_0$, where λ_0 is the wavelength the same line would have at rest in a lab. For velocities much smaller than c ,

$$z \approx \frac{v}{c},$$

with the convention that positive v means the source is receding (redshift, $z > 0$) and negative v means it's approaching (blueshift, $z < 0$). This is how astronomers measure a star's velocity along our line of sight without ever needing to watch it move — just by seeing how far its absorption lines have drifted from their known rest wavelengths.

Worked Example

A star's H α line, normally at $\lambda_0 = 656.3$ nm, is observed at $\lambda = 656.5$ nm. Find its radial velocity.

$$z = \frac{656.5 - 656.3}{656.3} \approx 3.05 \times 10^{-4} \implies v \approx zc \approx 91 \text{ km/s},$$

receding from us, since $z > 0$.

4 Epilogue: Three Questions, Three Tools

Every star answers three questions with the light it sends us, and this handout is organized around exactly those three tools:

How hot and how big is it (blackbody laws)? How bright does it look, and how far away is it (magnitudes)? What is it made of, and how fast is it moving (spectroscopy)?

And underneath all three tools sits the same habit of mind you already used for $PV = nRT$: a short list of carefully measured empirical laws (Wien's, Stefan's, Pogson's, the Rydberg formula) turns

out to be enough to read an enormous amount off a single beam of starlight — its temperature and size, its distance, its composition, and its motion. Nothing here required rebuilding physics from scratch at every step; it required only trusting a short list of rules and following them wherever the algebra led.

5 Summary Sheet

Topic	Result
Wien's Displacement Law	$\lambda_{max}T = b$
Stefan's Law	$E = \sigma T^4$
Luminosity from T, R	$L = 4\pi R^2 \sigma T^4$
Flux (inverse square law)	$f = L/4\pi d^2$
Pogson's equation	$m_1 - m_2 = -2.5 \log_{10}(f_1/f_2)$
Distance modulus	$m - M = 5 \log_{10}(d/10 \text{ pc})$
Rydberg formula	$1/\lambda = R_\infty Z^2(1/n_1^2 - 1/n_2^2)$
Mass-luminosity relation	$L/L_\odot = (M/M_\odot)^{3.5}$
Redshift	$z = (\lambda - \lambda_0)/\lambda_0 \approx v/c$